Curriculum Vitae

Zhenghui Li

Personal Information

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Position

09/2026-07/2027 **Postdoctoral fellow**: Institute for Advanced Study.
Supervisor: Prof. Bhargav Bhatt.

Education

10/2022-06/2026 **Ph.D in Mathematics**
Sorbonne Université
Thesis: Syntomic cohomology and arithmetic applications
Thesis director: Prof. Wiesława Nizioł.

10/2020-06/2022 **Master in Mathematics** ALGANT Master Program
Université de Bordeaux, France (2021-2022)
Universität Regensburg, Germany (2020- 2021)
Thesis: The Fargues-Fontaine Curve.
Thesis directors: Prof. Denis Benois & Prof. Wiesława Nizioł

09/2016-06/2020 **Bachelor in Science**
Nanjing University, China
Thesis: On Taniyama-Shimura Conjecture.
Thesis director: Prof. Chuangxun Cheng

Articles

1 On p-torsions of geometric Brauer groups (with Yanshuai Qin), preprint arxiv 2406.19518. (to appear in J. Reine Angew. Math.)

2 Duality for arithmetic p-adic pro-étale cohomology of analytic spaces, preprint arxiv 2412.11786.

3 F-isocrystals of higher direct images of p-divisible groups (with Yanshuai Qin), preprint arxiv 2506.11736.

4 Locally analytic vectors in the completed cohomology of quaternionic Shimura curves (with Benchao Su, Zhixiang Wu), preprint arxiv 2601.13625.

Invited Talks

2024	Séminar Mathjuene: Poincaré duality for p-adic étale cohomology of rigid analytic varieties. Nanjing University: Poincaré duality for p-adic étale cohomology of Stein spaces and p-adic Langlands program.
2025	IMJ-PRG number theory seminar: Arithmetic duality of p-adic pro-étale cohomology of analytic spaces. Workshop on arithmetic $K3$ surfaces (Lausanne): Boundedness of the p-primary torsion of geometric Brauer groups. Bucharest Number Theory days: Duality for arithmetic p-adic pro-étale cohomology of analytic spaces. Arithmetic and Algebraic Geometry Week (Iasi): On p-torsions of geometric Brauer groups. Séminaire Variétés Rationnelles (Jussieu): On p-torsions of geometric Brauer groups. Morningside center of mathematics: Duality for arithmetic p-adic pro-étale cohomology of analytic spaces.
2026	Copenhagen number theory seminar: On p-torsions of geometric Brauer groups. Caen University number theory seminar: On p-torsions of geometric Brauer groups. Beijing University: Bernstein–Zelevinsky duality for equivariant vector bundles on Drinfeld spaces.

Seminar Reports

2023	IMJ-PRG preprint seminar: $\mathbb{Z}_p$-lattices in semistable Galois representations. (après Zijian Yao)
2024	Séminaire d'étude de la thèse d'Arnaud: The cuspidal case.
2025	IMJ-PRG preprint seminar: Berthelot's conjecture via homotopy theory. (après Veronika Ertl and Alberto Vezzani)

Teaching

2026	Beijing number theory summer school: Reading course on $\langle p$-adic Hodge theory for rigid analytic varieties$\rangle$.

Grants

10/2023-09/2025	'MathInParis': Marie Sklodowska-Curie cofunding.
01/2024-06/2026	'Group schemes, root systems, and related representations': NextGenerationEU and Romania's National Recovery and Resilience Plan.
09/2026-07/2027	Simons Foundation.

Referee work

International Mathematics Research Notices; Nagoya Mathematical Journal

Attended conferences

07/2026	A p-adic gather. (Beijing)
07/2026	Workshop on p-adic modular forms and the Fontaine–Mazur conjecture. (Hefei)
01-02/2026	Thematic month on Langlands program. (Marseille)
11/2025	Arithmetic Geometry in Shenzhen II. (Shenzhen)
10/2025	Workshop on Non-archimedean geometry. (Lille)
09/2025	Motives, L-values and Eisenstein series. (Regensburg)
09/2025	Arithmetic and Algebraic Geometry Week. (Iasi)
05/2025	Workshop on condensed maths and Arakelov geometry. (Strasbourg)
05/2025	Workshop on arithmetic $K3$ surfaces. (Lausanne)
03/2025	Annual Meeting: Simons Collaboration on Perfection in Algebra, Geometry and Topology. (NewYork)
12/2024	Workshop on analytic de Rham stacks. (Warsaw)
11/2024	Workshop on p-adic geometry. (Singapoo)
09/2024	Summer school on Shimura variety and Hodge theory. (Essen)
07/2024	Conference on K-theory and arithmetic. (Bedlewo)
06/2024	Workshop on p-adic Riemann-Hilbert correspondence. (Padova)
01/2024	Conference on p-adic Geometry. (Shenzhen)
10/2023	Conference for Rapaport's birthday. (Munster)
08/2023	Workshop on THH and prismatic cohomology. (Shenzhen)
08/2023	Conference on algebraic and arithmetic geometry. (Nanjing)
07/2023	Workshop on Vanishing of étale cohomology of Shimura varieties. (Bonn)
05/2023	Summer school on arithmetic Langlands and p-adic geometry. (Bonn)
03/2023	Workshop on condensed maths. (Marseille)